

5. BFS

Aim:

To implement the Breadth-First Search (BFS) algorithm using Python to traverse a graph level by level.

Objective :

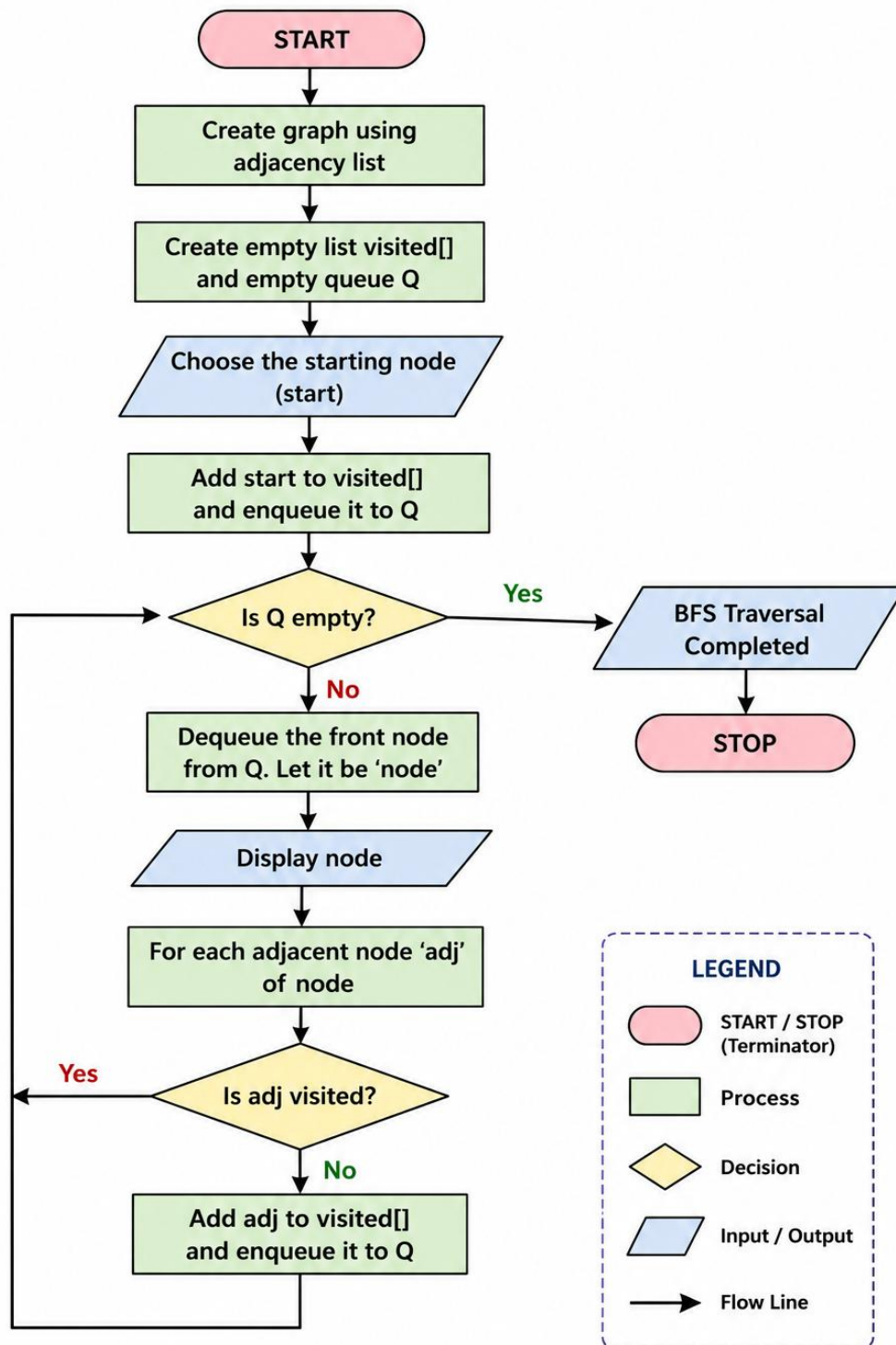
- To understand the Breadth-First Search (BFS) algorithm.
- To traverse all the vertices of a graph using a queue.
- To visit each node exactly once.
- To display the BFS traversal order.

Algorithm:

1. Start.
2. Create a graph using an adjacency list.
3. Create an empty visited list and an empty queue.
4. Choose the starting node.
5. Mark the starting node as visited and add it to the queue.
6. Repeat until the queue becomes empty:
 - Remove the first node from the queue.
 - Display the node.
 - Visit all its unvisited adjacent nodes.
 - Mark each adjacent node as visited and add it to the queue.
7. Stop.

Flowchart :

BREADTH-FIRST SEARCH (BFS) – FLOWCHART



Code:

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}  
visited = []  
queue = []  
start = 'A'  
visited.append(start)  
queue.append(start)  
print("BFS Traversal:")  
while queue:  
    node = queue.pop(0)  
    print(node, end=" ")  
    for neighbour in graph[node]:  
        if neighbour not in visited:  
            visited.append(neighbour)  
            queue.append(neighbour)
```

Output:

```
Output
BFS Traversal:
A B C D E F |
=== Code Execution Successful ===
```

Result :

The Breadth-First Search (BFS) algorithm was successfully implemented using Python. The graph was traversed level by level starting from the source node, and the traversal order is obtained.